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Informatica® RulePoint
6.2

Quick Start Guide for Amazon Web Services

Informatica RulePoint Quick Start Guide for Amazon Web Services
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Preface

CHAPTER 1

Getting Started with RulePoint Amazon Machine Image

This guide provides instructions to launch and connect to the RulePoint Amazon Machine Image (AMI). RulePoint AMI is configured with Elastic Block Storage (EBS) and contains RulePoint pre-installed on Linux. The database type in RulePoint AMI is H2.The RulePoint AMI contains a single-host, single-node topology, along with the system and application services. You cannot configure high availability for the topology components in the RulePoint AMI.

This guide also provides instructions to connect to an Amazon Kinesis stream, consume data, view activations, and publish metrics through a CloudWatch responder. The instructions help you have a basic understanding of how RulePoint works.

For more information about RulePoint objects, see the *RulePoint User Guide*. For instructions to manage and administer the topology, and import and export objects, see the *RulePoint Administrator Guide*. For sample use cases, see the *Banking Use Case Tutorial*, *Trading Use Case Example*, and *Business Process Management Example*.

CHAPTER 2

Before You Begin

Before you launch the RulePoint AMI, complete the following prerequisites:

1. Sign up for Amazon Web Services (AWS) and create an AWS account. For instructions, see the Amazon AWS documentation.
2. You must have a basic working knowledge of the Amazon Elastic Compute (EC2) environment. For more information, see the Amazon AWS documentation.
3. Subscribe to the RulePoint AMI and get the permissions to launch the AMI. To request the required permissions to launch the RulePoint AMI, contact your Informatica Sales Representative.

CHAPTER 3

Launching an EC2 Instance

Log in to the AWS Management Console with the email address and password that you used when you signed up for Amazon EC2. To use the RulePoint AMI, launch an EC2 instance. The RulePoint processes will take up to 5 minutes to start.

For more information about launching an instance, see the Amazon AWS documentation.

CHAPTER 4

Creating the EC2 Security Group

Select the security group type to control the type of network traffic that the EC2 instance accepts. The wizard in the AWS populates the launch-wizard-x security group to allow you to connect to the RulePoint instance.

The following table describes the ports that you can configure:

Type	Protocol	Port Range	Source
Custom TCP Rule	TCP	22	0.0.0.0./0
Custom TCP Rule	TCP	8080	0.0.0.0./0

Note: You can configure the subnet that you use in the Source column. The subnet 0.0.0.0/0 indicates that the source can be accessed from any IP address.

For more information about creating security groups, see the Amazon AWS documentation.

CHAPTER 5

Connecting to the RulePoint Amazon Machine Image

After you launch the RulePoint AMI, you can connect to it and use it. When you launch an instance and when it has started booting, the status of the instance is Running. The instance receives a public DNS name that you must copy and use to connect to the instance.

For more information about connecting to the RulePoint AMI, see Amazon AWS documentation.

1. To use the SSH client to connect to the RulePoint AMI, in a command line shell, change directories to the location of the private key file that you created when you launched the instance.
2. Type the Public DNS name or the IP address of the running RulePoint AMI.

   Note: You will find the public DNS name or IP address under **Instances** on the **Services** tab of the Amazon EC2 console. The public IP address will change every time you reboot the instance.

CHAPTER 6

Logging In to RulePoint

After the design-time and topology instances of RulePoint start, log in to the RulePoint console.

1. Launch a web browser and enter one of the following URLs:
   - `http://<Public IP address of the current EC2 session>:<port>/rulepoint`
   - `http://<Public DNS of the current EC2 session>:<port>/rulepoint`

   Note: The default port is 8080.

   For example:
   - http://10.10.10.10:8080/rulepoint
   - http://ec2-10-10-10-10.compute-1.amazonaws.com:8080/rulepoint

   The RulePoint login page appears.
2. Enter the RulePoint login credentials.

   The user name is Administrator. The password is the EC2 instance ID of the current AMI session.

   Optionally, change the password after logging in.
3. Click **Log In**.

   The Informatica RulePoint home page appears.

CHAPTER 7

Creating a Topic for the Amazon Kinesis Source

Before you create a Kinesis source, you must configure a topic for the Kinesis source. For example, create a topic, aws_topic, for the types and properties of events that come into RulePoint from Amazon Kinesis.

1. On the **Design** tab, click the **Topics** view.
2. From the **Actions** menu in the upper-right corner of the view, select **New**.

   The **New Topic** dialog box appears.
3. In the **Details** section, enter a name and description for the topic.
4. In the **Properties** section, add the properties that you want the topic to pick up from the Kinesis stream.

   For example, you can add the following property names:

   - aktimestamp
   - akpartitionkey
   - aksequencenumber
   - akmsg
5. Click **Add**, and then click **Save**.

CHAPTER 8

Creating an Amazon Web Services Connection

Create an Amazon Web Services connection so that the Kinesis source in RulePoint connects to a stream in Amazon Kinesis. You can also use the connection in the CloudWatch responder in RulePoint to connect to Amazon CloudWatch.

1. On the **Design** tab, click the **Connections** view.
2. From the **Actions** menu in the upper-right corner of the view, select **New**.

   The **New Connection** dialog box appears.
3. Enter a name and description for the connection.
4. Select the type **Amazon Web Services Connection**.

   Note: If you specify an IAM role while launching the EC2 instance, the values for the **Access Key ID** and **Secret Access Key** parameters are autopopulated. If you do not specify an IAM role for the EC2 instance, manually enter the values for the parameters.
5. Click **Save**.

CHAPTER 9

Creating an Amazon Kinesis Source

Create a Kinesis source that uses an Amazon Web Services connection to connect to Amazon Kinesis.

1. On the **Design** tab, click the **Sources** view.
2. From the **Actions** menu on the upper-right corner of the view, select **New**.

   The **New Source** dialog box appears.
3. Enter a name for the Kinesis source, and select the type **Amazon Kinesis Source**.
4. Select the configured Amazon Web Services connection.
5. In the **Topic Information** section, select the configured topic for the Kinesis source.
6. In the **Configuration** section, configure the following parameters:
   a. Enter a stream name to identify the stream that you want to read from Amazon Kinesis.
   b. Select the **Latest** shard iterator type to read the most recent data in the shard.
   c. Select the **String Message** message type.

      Note: Select JSON encoded message if you want to parse JSON messages and publish events with key-value pairs. Select custom encoding message type if you want RulePoint to process other message types. For custom encoding message type, you need a marshaler. The JSON encoded or string message type processes a single event at a time. You must create a marshaler and use the custom message encoding type to process multiple events at a time.

      Retain the default values for the rest of the fields. For information about the description for each of the fields, the configurations for the Kinesis source and its associated objects, and marshaler details, see the *RulePoint User Guide*.
7. Click **Save**.

   The created source is in Draft state.

CHAPTER 10

Creating a CloudWatch Responder and Response

The responder requires an Amazon Web Services connection to connect to Amazon CloudWatch and write messages to it. After you create the responder, create a response by specifying the parameters that you want to write to CloudWatch.

1. On the **Design** tab, click the **Responders** view.
2. From the **Actions** menu in the upper-right corner of the view, select **New**.

   The **New Responder** dialog box appears.
3. Enter a name for the responder, select the type Amazon CloudWatch responder, and select the configured Amazon Web Services connection.
4. In the **Configuration** section, configure the following parameters:
   - Namespace, for example, the string defined for the metric you want to publish to CloudWatch.
   - Metric name, for example, Test_Metric, to identify the metric.
   - Dimensions, for example, TestDim=TestVal. This is a name/value pair that provides additional information to identify a metric.
   - The value for the metric.
   - The unit for the metric.

   For more information about each of these parameters, see the *RulePoint User Guide*.
5. Click **Save**.

   The responder is in Draft state. If you do not create a response associated with the responder, the responder is not in a valid state. You must create a response and associate the response to the responder before you deploy the responder.
6. From the **Responses** view on the **Design** tab, select **New** from the **Actions** menu.

   The **New Response** dialog box appears.
7. Enter a name for the response, and then select the responder type that you created.
8. Click **OK**, and then click **Save**.

CHAPTER 11

Creating a Rule

Create a wizard rule that contains information about the topic, the number of events to process, the conditions to check, and the response to generate when the event meets the specified condition.

To create an advanced rule or a template rule, see the *RulePoint User Guide*.

1. On the **Design** tab, click the **Rules** view.
2. From the **Actions** menu in the upper-right corner of the view, select **New Wizard Rule**.

   The **New Wizard Rule** dialog box appears.
3. In the **Details** section, enter the name and description for the rule.
4. In the **Topics and Events** section, click **Add Topics**, and then select the configured topic that you want to process, and click **OK**.
5. In the **Events Processed at a Time** column, specify how many events you want to process at a time.

   If you specify a count greater than 1, additional options appear, and you must specify how you want the events processed.
6. In the **Conditions** section, click **Add Conditions**.
7. Enter the condition name, and then select the condition type from the available options.

   For example, if you use the **Compare topic property to a literal** option, select the topic property, operator, and the value with which you want to compare it. Use this rule to find out whether you can search if a certain string message or number exists in the data stream. For more information about conditions and operators, see the *RulePoint User Guide*.
8. Click **Add**.
9. In the **Responses** section, select the configured response for the rule.
10. Click **Save**.

    The wizard rule is in Draft state. You must deploy the wizard rule.

CHAPTER 12

Deploying the Objects

When you deploy the objects, the primary objects along with their supporting objects are deployed in the default application services for further processing. The Kinesis source with the associated topic and connection are deployed in the source controller. The CloudWatch responder and the connection are deployed in the responder controller. The rule is deployed in the event processor.

1. On the **Design** tab, click the **Projects** view.
2. From the navigator **Actions** menu, click **Deploy** > **Rules, Sources & Responders**.
3. In the **Deploy Rules, Sources & Responders** dialog box, perform the following tasks:
    a. Select the wizard rule that you want to deploy, and click **Next**.
    b. Select the source that you want to deploy, and click **Next**.
    c. Select the responder that you want to deploy, and click **Deploy**.

CHAPTER 13

Viewing Events, Activations, and Alerts

The Kinesis source reads messages from the Amazon Kinesis stream and publishes the messages as events to a topic. When an event matches a rule condition, the responder sends an alert to the configured user or system. You can view the statistics of the number of events, activations, and alerts on the **Metrics** view of the dashboard.

1. Click the **Dashboard** tab.
2. In the **Metrics** view, select the time line for which you want the statistics for the deployed objects.
3. To view the statistics for the Kinesis source, select the source controller on the left pane.

   The contents panel displays the number of events generated for the source.
4. To view the statistics of the deployed rule, click the event processor on the left pane.

   The contents panel displays the number of rule activations.
5. To view the statistics of the CloudWatch responder, click the responder controller on the left pane.

   The contents panel displays the number of response activations.

   Note: The **Events** and **Logs** view are disabled.

CHAPTER 14

Starting RulePoint Manually

When you connect to RulePoint, the components such as the design-time and topology instances are started. If there is a requirement where you need to start RulePoint manually, use the SSH client to start the database and the RulePoint components. If you start the RulePoint instances manually, you must set the RULEPOINT_HOME environmental variable.

1. To set RULEPOINT_HOME to the location of the `mnt\apps\RulePoint_6.2` directory, run the following command:

   ```
   $export RULEPOINT_HOME=\mnt\apps\RulePoint_6.2
   ```

2. Start the RulePoint instances from the `\mnt\apps\RulePoint_6.2\bin` directory. Perform the following tasks:

   a. To start the design-time instance, run the following command:

      ```
      design.sh start
      ```

   b. To start the RulePoint topology instance, run the following command:

      ```
      topology.sh start Default
      ```

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